

EE 210L — Digital Circuits and Logic Design Lab

Lab 8: Combinational Logic in Verilog

Name: _____ Date: _____

Partner(s): _____ Section: _____

*Paste your source code and waveform captures in the boxes provided. Simulation only — no hardware this week.
Submit this report on Canvas.*

Part 1 — Majority Voter

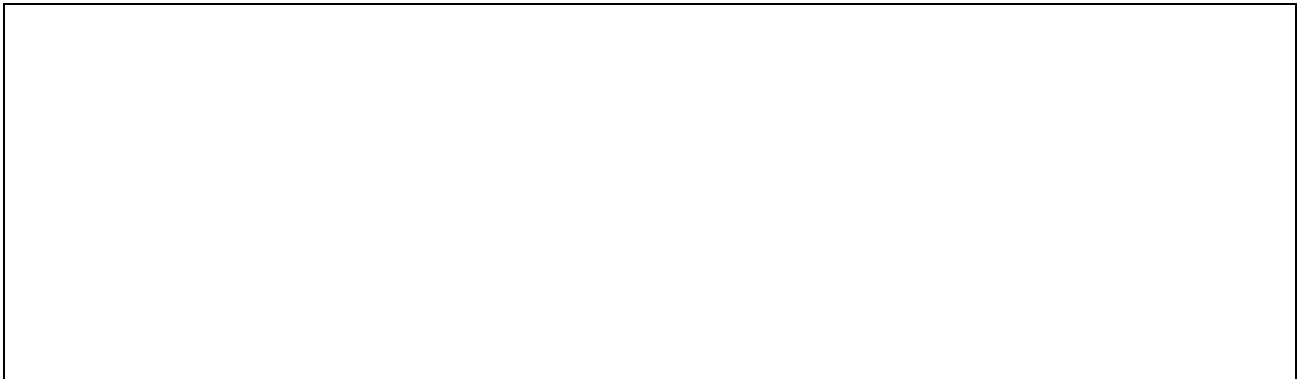


Figure 1: majority.v source.



Figure 2: Simulation waveform, all eight cases.

Testbench console output: _____

Part 2 — Break It on Purpose

With the `a` & `c` term removed, the testbench printed:

Which input case(s) failed? _____

How would you have found this on the waveform alone?

Part 3 — Two-Bit Comparator

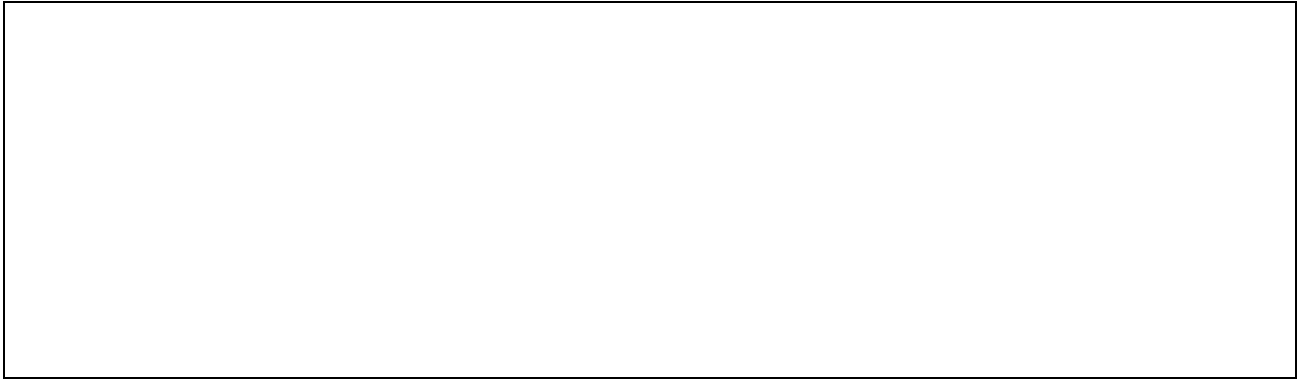


Figure 3: compare2.v source.

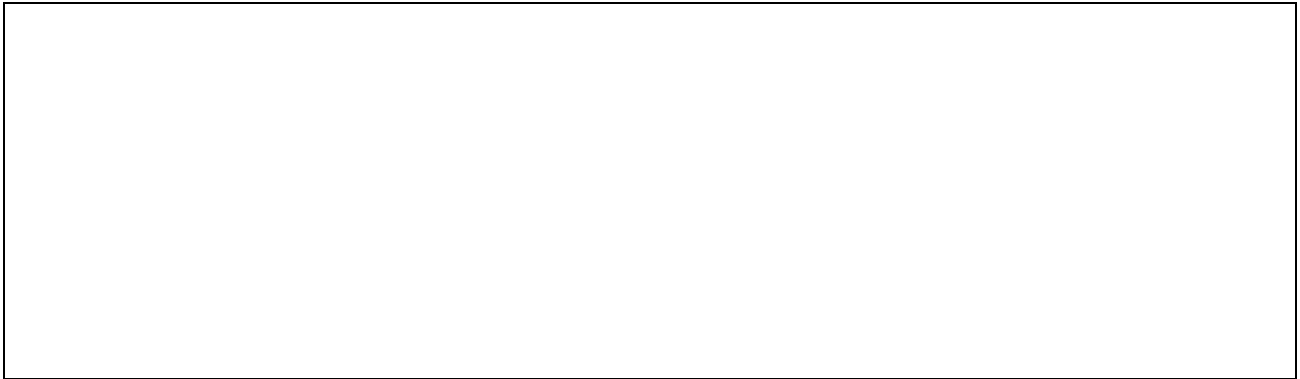


Figure 4: Simulation waveform, all 16 cases.

All 16 cases passed? _____ Did you add the “exactly one output high” check?

What class of bug does that extra check catch that testing gt, eq, and lt separately would miss?

Part 4 — Inferred Latch

Copy the synthesis warning verbatim.

After assigning a default, was the warning gone? _____

Explain why the original code required a latch.

Part 5 — Four-Bit Adder

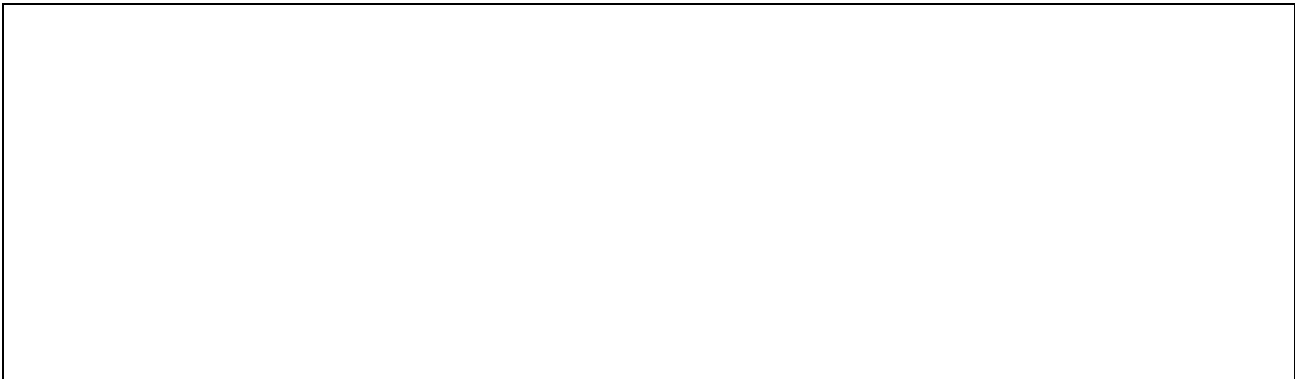


Figure 5: adder4.v and testbench source.

Cases verified	Result (pass / fail)	Approx. time to write

Compare this with wiring and measuring the 74LS283 in Lab 5. How many input combinations did you check by hand there, versus here?

Discussion and Conclusions

Address at least: what "Verilog describes hardware, it does not execute" means in practice, why a self-checking testbench beats reading waveforms, and how accidental latches happen.
